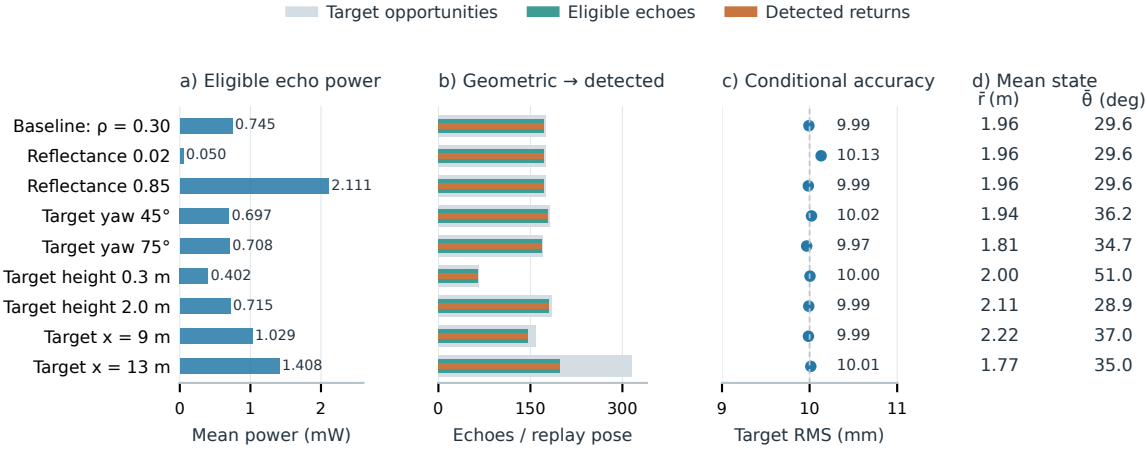

Nine target conditions · 27 replays of the same three saved baseline trajectories (271 poses per condition).

RMS is pooled over detected target returns; eligible power/range/incidence are conditioned on geometric echoes within range gates.

Position labels denote target world x, not a fixed sensing distance. Whole-route means mix changing range and incidence.

Baseline target: x = 11 m, height 1.2 m, yaw 0°, diffuse reflectance 0.30. Other robot, sensor, and weather settings are fixed.